

DP-800

Develop AI-enabled database solutions

Trainer Preparation Guide

March 2026

Purpose

This document is for Microsoft Certified Trainers preparing to teach DP-800. This course is designed for students who are planning to take the corresponding certification exam, or students who are performing database related tasks in their daily job.

Course description

This course teaches developers how to design, build, secure, and deploy AI-enabled database solutions using Microsoft SQL platforms. Students learn advanced T-SQL techniques and modern data patterns for working with structured and semi-structured data, and how to integrate AI capabilities, such as embeddings, vectors, semantic search, and retrieval-augmented generation (RAG), directly within SQL-based applications. The course also covers best practices for performance optimization, security, and enterprise deployment, with an emphasis on CI/CD workflows and collaboration across application, database, and DevSecOps teams.

Learning objectives

After completing this course, students will be able to:

- Design and build database solutions using structured and semi structured data.
- Integrate AI features into modern, scalable applications.
- Secure, optimize, and deploy enterprise grade SQL solutions.
- Implement AI capabilities directly within database architectures.

SQL AI Developer Associate role definition

Both the certification exam and the courseware are based on the SQL AI Developer Associate role.

Your responsibilities include:

- Designing and developing database solutions that include both structured and semi-structured data.
- Integrating AI features into modern and highly scalable enterprise applications.
- Securing, optimizing, and deploying database solutions.
- Implementing AI capabilities in database solutions.

You work closely with application developers; database administrators (DBAs); architects; AI engineers; development, security, operations (DevSecOps) engineers; security and compliance administrators; and other stakeholders to deliver robust, high-performance database solutions that power modern applications and AI-driven experiences.

Audience profile

As a candidate for this Microsoft Certification, you should have subject matter expertise in designing and developing AI-enabled database solutions across Microsoft SQL platforms, including Microsoft SQL Server, Azure SQL, and SQL databases in Microsoft Fabric.

Audience prerequisites

You should have experience writing T-SQL code and developing databases in Microsoft SQL platforms. Plus, you need to be familiar with continuous integration and continuous deployment (CI/CD) practices in GitHub, AI-assisted development tools, and AI concepts, such as embeddings, vectors, and models.

Certification exam

The study areas for the certification exam related to this course are based on the Job Task Analysis (JTA) that was conducted in December 2025.

Each study area has a percentage indicating the relative weight of the area on the exam. The higher the percentage, the more questions a candidate is likely to see in that area.

To review the full objective domain the skills measured, review the [DP-800 exam page](#).

Study Area	Percentage
Design and develop database solutions	35-40%
Secure, optimize, and deploy database solutions	35-40%
Implement AI capabilities in database solutions	25-30%

For exams related to non-Fundamental courses, candidates should have a minimum of six months of hands-on experience working with SQL Server, Azure SQL Database or SQL database in Fabric.

Candidates should understand core Azure services, Azure workloads, security, and governance. Candidates for this exam should have experience using PowerShell, the Command Line Interface, and the Azure Portal.

Prerequisite knowledge to teach this course

To successfully teach this course, instructors must have hands-on experience designing and building SQL-based database solutions and integrating AI capabilities into applications that use Microsoft SQL platforms.

Note: These guidelines are not inclusive of the requirements to become a Microsoft Certified Trainer (MCT).

- Advanced T-SQL skills (query design, window functions, error handling, performance tuning, indexing basics).
- Experience with Microsoft SQL platforms such as SQL Server and Azure SQL, and familiarity with SQL databases in Microsoft Fabric.
- Working understanding of AI concepts used in data solutions, including embeddings, vectors, similarity search, and model fundamentals.
- Ability to explain and implement RAG patterns at a high level, including grounding techniques and strategies to reduce hallucinations.
- Familiarity with securing and deploying database solutions, including identity/permissions concepts and basic compliance considerations.
- Comfort with CI/CD workflows in GitHub for database or application projects (branching, pull requests, pipelines).

Required materials to prepare for and teach this course

You need the following materials to prepare for and teach this course:

Resource	Description
Microsoft PowerPoint files	Download the DP-800T00-ENU-PowerPoint.zip from the MCT Download Center .
Change Log	Download the DP-800T00-ENU-ChangeLog.pdf from the MCT Download Center .
Lab environment provided by your lab hosting provider	Contact your lab hosting provider for instructions on using their lab environment.
Lab instructions	The lab instructions are provided in the lab environment and in the DP-800 Microsoft Learning GitHub repository SQL Developer Exercises .
Student training content	See the following section for a detailed breakdown of each Learning Path covered in the course.
Microsoft Learn for Educators program	For educational institutions that deliver this course, additional assets are available through the Learning Download Center (LDC) for Educators.

Student training content

The student training content for this course is located in Microsoft Learn. The following table provides a breakdown of each Learning Path, the modules covered in each, and the link to each LP in Microsoft Learn.

The student training content includes links to additional reading material to help you prepare for specific topic areas.

Learning Path	Online training in Microsoft Learn
Course Introduction	Slides only
Learning Path 1: Design and develop database solutions	Module 1: Design and implement database objects with SQL Lab: Design and implement database objects with SQL Module 2: Implement programmability objects with SQL Lab: Implement programmability objects with SQL Module 3: Write advanced T-SQL code Lab: Write advanced T-SQL queries Module 4: Implement SQL solutions by using AI-assisted tools Lab: Implement SQL solutions by using AI-assisted tools
Learning Path 2: Secure, optimize, and deploy database solutions	Module 1: Implement data security and compliance with SQL Lab: Implement security and compliance with SQL Module 2: Optimize database performance

	Lab: Optimize query performance Module 3: Implement CI/CD by using SQL Database Projects Lab: Implement CI/CD with SQL Database Projects Module 4: Integrate SQL solutions with Azure services Lab: Integrate SQL solutions with Azure services
Learning Path 3: Implement AI capabilities in database solutions	Module 1: Design and implement intelligent search with SQL Lab: Generate and update embeddings in Azure SQL Database Module 2: Design and implement models and embeddings with SQL Lab: Implement intelligent search with full-text, vector, and hybrid queries Module 3: Design and implement RAG with SQL Lab: Implement RAG solutions with SQL

Preparation tasks

Instructors should complete the following tasks to prepare for teaching this course:

- If you have previously taught this class, refer to the course's Change Log. It provides detailed information on how the course has changed over time. The Change Log is updated for each course release.
- Review all topics in the student training material in Microsoft Learn (see the link in the Required Materials section above). You should be well-versed in every topic. If you have previously taught the course and are comfortable with your knowledge of each topic, focus primarily on the new or updated topics as outlined in the Change Log.
- Review the PowerPoint slides.
 - Be able to speak to each of the talking points on the slides. Some slides include a graphic from the associated Learn content for the topic. These graphics are provided on the slide so that you can speak to them to help explain the key talking points in the topic.
 - **The bulleted items on each slide should NOT be read verbatim to the students.** The students can read the slides themselves. Rather, the bullet points reflect the key information that you should focus on when discussing each topic. You should use your experience as a subject matter expert to explain the What, the Why, and the How of each topic. **This is your opportunity to provide a real value-add above and beyond the bulleted talking points.**
- Review the Additional Reading links and other linked resources provided in the student training material. **It's recommended that you present key points from this material to supplement the value-add you provide as an instructor.**
- As you prepare for the class, you should review each unit and determine which ones you want to perform demonstrations of the corresponding product functionality. It's up to you to decide which

product features you want to demonstrate to the class. You should use your experience to identify key points during the demonstration process. **This is an area where you should rely on your experience as a subject matter expert to provide additional value-add to the students.**

- You should review each Knowledge Check (KC) question so that you know why the correct answer is correct for each question. Students may challenge some of the questions, so you must be able to address any of those concerns.
- You should perform the labs yourself prior to class so that you become familiar with them and with any of the difficult points in the lab exercises. This will prepare you for helping students in case they get stuck.

Course timing

Daily agenda

The following agenda provides estimated times to complete each classroom activity. However, the estimated times may vary depending on the background of your students, which may affect whether you can move faster or slower through the course material.

Estimated times for each Module include the time to complete as part of a 3 day course:

- The module's PowerPoint slide deck presentation.
- Any pre-defined product demonstrations.
- Time to review Knowledge Check questions (see the section on Additional Timing Notes below).
- Time to complete a classroom discussion activity if a Discussion slide is included in the module slide deck.

For courses taught in a longer format such as part of the Microsoft Learn for Educators (MSLE) program, review the **Design your syllabus** section of the **Educator Teaching Guide** found on the [Learning Download Center](#).

You should adjust the agenda accordingly based on any classroom activities that you personally created or plan to deliver that are not included in the slides for this course. For example, if you plan to present:

- ad-hoc demonstrations
- review activities
- classroom games
- and so on...

Note: Each Learning Path/Module activity in the following agenda is the slide deck presentation for that module.

The following table provides delivery specifications for a 3-day course. For courses taught in a longer format such as part of the MSLE program, review the **Design your syllabus** section of the **Educator Teaching Guide** found on the [Learning Download Center](#).

Day	Estimated Time	Classroom activity
1		Learning Path 1: Design and develop database solutions
	50 min	(LP1, M1) Design and Implement Database Objects
	25 min	Exercise: Create Database Objects
	50 min	(LP1, M2) Implement Programmability Objects
	55 min	Exercise: Implement Programmability Objects
	55 min	(LP1, M3) Write Advanced SQL Code
	40 min	Exercise: Write Advanced SQL Code
	60 min	(LP1, M4) Design and Implement SQL Solutions with AI-Assisted Tools
	25 min	Exercise: AI-Assisted SQL Development
2		Learning Path 2: Secure, optimize, and deploy database solutions
	70 min	(LP2, M1) Implement Data Security and Compliance
	40 min	Exercise: Implement Security Features
	70 min	(LP2, M2) Optimize Database Performance
	55 min	Exercise: Optimize Query Performance
	70 min	(LP2, M3) Implement CI/CD for SQL Database Projects
	55 min	Exercise: Implement CI/CD for SQL Database Projects
3	45 min	(LP2, M4) Integrate SQL Solutions with Azure Services
	40 min	Exercise: Data API Builder
		Learning Path 3: Implement AI capabilities in database solutions
	50 min	(LP3, M1) Design and Implement Models and Embeddings with SQL
	30 min	Exercise: Generate and Update Embeddings
	45 min	(LP3, M2) Design and Implement Intelligent Search with SQL
	40 min	Exercise: Implement Intelligent Search
	50 min	(LP3, M3) Design and Implement RAG with SQL
	40 min	Exercise: Implement a RAG Solution
	15 min	Course Q&A and Wrap-Up

Additional timing notes - Knowledge check questions

Knowledge check (KC) questions are provided throughout the course to check the student's knowledge of the material that was covered. Instructors can use these KC questions in several ways:

- Conduct a formal classroom exercise in which you go through the questions in a module before moving on to the next module.
- Sprinkle the questions into the content as you cover the related material for a module
- Let the students review the questions after class as a daily homework assignment. You can set aside time at the start of each day to answer any questions they have regarding the prior day's questions. This may be the most feasible option given the tight time constraints that most classes work under.

It will be left up to each instructor to determine how they want to incorporate the KC questions into their class.

If you provide students with time to review the KC questions at the end of specific topics and at the end of each module, you should provide a couple of minutes per question, along with a few extra minutes per question to respond to student questions or challenges concerning certain questions they may not understand or whose answers they disagree with. This may add an extra 15 to 30 minutes to complete each module.

Labs

The labs must be completed within the lab environment provided by your lab hosting provider. Detailed, step-by-step instructions are provided for each lab and presented as part of the UI experience within your lab environment.

At the time the courses were released, the lab instruction had been thoroughly tested and the lab steps were 100% accurate. However, given the nature of Microsoft's cloud products and the fact that Microsoft releases UI updates on a regular basis, it's possible that at some point in time, the UI for a given feature may change so that it no longer matches the lab instruction.

If students encounter lab steps that don't accurately reflect the UI, they'll have to work through the UI to determine what needs to be done. Typically, UI changes are quite subtle, so hopefully you don't find yourself in a situation where a feature was completely overhauled.

However, if you do run into major UI changes, challenge your students to work through it, and only offer help if they definitely need it. Product UI changes will be part of their daily life in today's cloud-centric world. As IT/Pros, they must learn how to work through such situations.

One thing Microsoft does ask of you is that if you run into situations such as this where lab instructions no longer match the corresponding UI, please document the issue in the course's GitHub repository. This will help Microsoft's World-Wide Learning team update the lab instructions to keep them as up to date as possible. For information on how to submit an issue, please see [GitHub User Guide for MCTs](#).

Common product misconceptions

Below are some common misconceptions about the SQL products:

- **"sp_invoke_external_rest_endpoint only works in Azure SQL Database."**
This stored procedure is available in SQL Server 2025, Azure SQL Managed Instance, Azure SQL Database, and SQL database in Microsoft Fabric. In Azure SQL Database it is enabled by default; in SQL Server 2025 and Azure SQL MI it is disabled by default and must be turned on with `sp_configure`. The real Azure SQL-only feature is the *allowed endpoints* approved lists, which restricts outbound calls to approved Azure services. Teams that assume this feature is Azure-exclusive may miss opportunities to call REST APIs directly from SQL Server 2025 without middleware.

- **"Vector embeddings are just another index type in SQL Server."**
Embeddings are stored in columns using the VECTOR data type (introduced in SQL Server 2025 and Azure SQL Database). A vector index such as a DiskANN index is a separate structure built on top of that column to accelerate approximate nearest-neighbor (ANN) searches. Confusing the data type with the index leads to poor schema decisions and missed performance tuning for similarity search workloads.
- **"If you add AI to your queries, the database schema does not matter much."**
RAG solutions depend on clean, well-normalized schemas with clear naming, referential integrity, and appropriate data types. The language model can only ground its answers on the data it retrieves, so poorly structured source data produces unreliable results regardless of how powerful the model is. Schema design is the single biggest factor in RAG answer quality.
- **"Always Encrypted and Transparent Data Encryption do the same thing."**
TDE encrypts the database files at rest and is transparent to every query. Always Encrypted keeps encryption keys outside the database engine so that neither DBAs nor server-side code can see plaintext. TDE protects against stolen disk media; Always Encrypted protects against anyone who has database access but should not see sensitive column values. Choosing the wrong option can leave a significant gap in the data-protection strategy.
- **"Dynamic Data Masking is a security control that stops unauthorized users from reading data."**
Microsoft's own documentation states that DDM does not prevent direct query access to the underlying values. Users with sufficient privileges can still retrieve unmasked data. DDM reduces casual exposure in application output, but it is not a substitute for column-level encryption or row-level security when true data protection is required.
- **"GitHub Copilot in Azure Data Studio writes production-ready T-SQL, so I can skip code review."**
Copilot is a productivity aid that generates suggestions based on patterns, not a substitute for human review. It may produce syntactically correct SQL that is logically wrong, ignores security policies, or performs poorly at scale. Every AI-generated query should be reviewed for correctness, performance implications, and compliance with organizational standards before running in production.
- **"Database CI/CD works the same as application CI/CD."**
Application CI/CD typically deploys immutable build artifacts. SQL database CI/CD is state based: tools like SqlPackage compare a .dacpac (a snapshot of the desired schema) against the live database and generate an incremental deployment plan that applies only the differences. Understanding this distinction is critical because rolling back a schema change is far harder than redeploying a prior application build.
- **"Data API Builder replaces the need for a backend API."**
Data API Builder generates REST and GraphQL endpoints from configuration, which is excellent for standard CRUD operations. However, it does not support custom business logic, multi-step transactions, or complex authorization flows. For scenarios that go beyond simple data access, a purpose-built backend API is still required. DAB is best positioned as a fast-start layer that sits alongside, instead of, a full API tier.

Real-world use cases

Some common real-world scenarios using Azure SQL Database or SQL Server 2025 features are:

- **Customer Support Chatbot with RAG**

A support organization stores its knowledge-base articles in Azure SQL Database with vector embeddings generated by Azure OpenAI. When a customer asks a question, the application converts the question into an embedding, runs a vector-similarity search to find the most relevant articles, then sends those articles as grounding context to an LLM. The model generates a natural-language answer rooted in verified documentation rather than general training data. This architecture combines the reliability of structured SQL storage with the conversational quality of a large language model.

- **Schema Drift Detection with SchemaCompare**

A team managing dozens of Azure SQL databases uses SQL Database Projects and SchemaCompare to detect unplanned schema changes. Each database is periodically compared to the canonical .dacpac stored in source control. SchemaCompare presents every difference (added columns, changed data types, dropped indexes) and lets the team accept or reject each one. SqlPackage DriftReport can also generate an XML report of changes applied since the database was last registered. This approach catches "Friday-afternoon fixes" that bypass the normal CI/CD pipeline.

- **Row-Level Security for Multi-Tenant SaaS**

A SaaS vendor hosts all tenants in a single Azure SQL Database. A security policy uses SESSION_CONTEXT to filter rows, so each tenant sees only its own data. The application sets the TenantId value in SESSION_CONTEXT at connection time, and the RLS predicate function transparently filters every query. This eliminates the need for WHERE TenantId = @id in every stored procedure and removes an entire class of data-leakage bugs.

- **Query Store Performance Triage**

- After a production deployment causes regression, a DBA opens the Query Store Regressed Queries view to identify plans that have degraded. Query Store retains multiple execution plans per query, and the DBA uses plan forcing (sp_query_store_force_plan) to revert a problematic query to its previous fast plan without any application code changes. Wait-statistics tracking (available since SQL Server 2017) reveals whether the bottleneck is CPU, memory, or IO, and top-N

query analysis pinpoints the most resource-intensive statements for optimization. All of this is built in and requires no third-party tooling.

- **MCP Server for AI-Assisted Database Development**

- A development team configures an MCP (Model Context Protocol) server for SQL Server inside VS Code or another AI-enabled editor. The MCP server exposes tools for listing tables, describing schemas, analyzing indexes, extracting stored-procedure definitions, and running read-only queries. The AI assistant uses these tools to answer natural-language questions about the database ("Which tables reference the Customers table?" or "Find missing indexes for the Orders schema") without the developer writing any exploratory SQL manually. Packages like mssql-mcp-server provide 29 database-analysis tools out of the box, and Microsoft has published its own MCP server repository for Azure SQL and SQL Server.

- **Event-Driven Data Sync with Change Data Capture**

- A retail company enables CDC on its Orders table in Azure SQL Managed Instance. CDC reads the transaction log and records every insert, update, and deletes into mirrored change tables with metadata columns (operation type, commit LSN, update mask). A downstream ETL process polls the CDC functions (fn_cdc_get_net_changes) on a schedule, picks up only the rows that changed since the last poll, and pushes them to a data warehouse. This avoids full-table scans and dramatically reduces both sync latency and load on the source system compared to traditional timestamp-based polling.

Feedback

In this course, we have provided a framework for you to work with. Take time to prepare and think about the value that only an instructor can bring to training. We hope to partner with you to provide an exceptional student experience, and we welcome your feedback.